

S V V Sai Sri Vallabh Pataneni

Research Assistant/Associate (PhD): Fuel Cell Systems, PEM & SOFC

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SUMMARY

Automotive Engineering Master's student at RWTH Aachen with two years of hands-on **PEM fuel-cell systems** experience, spanning a student-competition stack, a 100 kW automotive FCEV at Robert Bosch, and a 150-200 kW aircraft powertrain at DLR. At DLR he designed and built the complete **cathode-side control architecture**, from a 0D lumped-parameter plant model through feed-forward, PI and **Model Predictive Control (MPC)**, now calibrated against real measurement data. He leads fuel-cell testing on **Germany's only student-built hydrogen vehicle**, where his energy-management strategy already lifted efficiency by roughly 80% for a runner-up finish at Shell Eco-Marathon 2025, and co-authored a rule-based FCEV control paper under review at SAE. He is applying to RWTH's **Chair of Thermodynamics of Mobile Energy Conversion Systems (TME)** to bring this system-modeling, control and test-bench experience to fuel-cell systems research for mobile applications.

PROFESSIONAL EXPERIENCE

Werkstudent, Fuel-Cell Systems (Aircraft) | DLR Institute for Small Aircraft Technologies (INK) 01/2026 – Present

Würselen (Aachen), Germany

Sized and selected the cathode, thermal-management and sensory balance-of-plant for the 150-200 kW EnginePod fuel-cell system powering DLR's hydrogen-electric 9-seater commuter aircraft, building Python- and Excel-based sizing tools to derive the design specifications and driving procurement. Designed the full system rig in CATIA V5 for packaging, piping, testing and component orientation. On that basis, built the complete cathode-side control architecture in MATLAB/Simulink: a 0D lumped-parameter plant model driving feed-forward, PI, decoupler and Model Predictive Control (MPC) loops, now calibrated against real measurement data, with a higher-fidelity 1D model underway in GT Suite.

Key Skills: MATLAB/Simulink, Model Predictive Control, GT Suite, Balance-of-Plant Sizing.

Mechanical / Suspension Team Lead & Pit Crew | Ecogenium e.V. (RWTH student team) 06/2024 – Present

Aachen, Germany

Leads fuel-cell system testing, integration and fault diagnosis on Germany's only student-built hydrogen fuel-cell vehicle, and wrote the energy-management and cooling-control strategy for its new water-cooled stack in MATLAB/Simulink. Previously engineered, fabricated and wired a standalone test bench for the team's air-cooled stack, running purging, humidity-cycling, short-circuiting and polarisation-curve campaigns to map its thermal operating window. That end-to-end ownership, from component sizing and CAN-bus communication through control design to on-track testing, lifted fuel efficiency by roughly 80% and secured a runner-up finish at Shell Eco-Marathon 2025.

Key Skills: Fuel-Cell Test-Bench Design, Energy Management Strategy, CAN Communication, Team Leadership.

Operating Strategy / System Simulation Intern | Robert Bosch GmbH 09/2025 – 12/2025

Schwieberdingen, Germany

Engineered a MATLAB-based powertrain strategy for strong-hybrid FCEVs that safeguards fuel-cell and HV-battery degradation, managing dynamic loads, thermal stress, low-voltage operation, SOC swing and life cycles through MAPs parameterized for different battery capacities and drive cycles. Developed and validated an iterative Simulink model of a rule-based power-split operating strategy in the (FC)EVsim PHEV module, correlating outputs against reference Excel and measurement data. Executed multi-variable parametric optimization with Genetic Algorithms and Gradient Descent to generate control MAPs, benchmarked against Dynamic Programming and ECMS, and formalized the resulting KPI analysis, covering power distribution, SOC window, efficiency and degradation, into a peer-reviewed manuscript now in review at SAE.

Key Skills: Genetic Algorithms, Gradient Descent, DP/ECMS Benchmarking, Degradation Mitigation.

SKILLS

Fuel-cell systems & controls

PEM fuel-cell system architecture

0D lumped-parameter plant modeling

1D lumped modeling

feed-forward/PI/decoupler control design

Model Predictive Control (MPC)

back-pressure-valve (BPV) protection control

operating-strategy optimization

balance-of-plant sizing & selection

CAN communication

Modeling & simulation

MATLAB/Simulink

GT Suite

Dymola (Modelica)

Genetic Algorithms

Gradient Descent

DP/ECMS benchmarking

Test & measurement

fuel-cell test-bench build & commissioning

single-cell & stack testing

polarisation curves

purging/humidity cycling

measurement-data evaluation & troubleshooting

Programming & data

Python (NumPy, Pandas)

C/C++

Power BI

Design & CFD

CATIA V5

Siemens NX

ANSYS (Fluent, Structural)

EDUCATION

M.Sc. Automotive Engineering, RWTH Aachen University

09/2023 – present

Aachen, Germany · GPA 1.7 (1.2 in core automotive courses) · Master's thesis planned at DLR on fuel-cell system control ·

Relevant modules: Alternative & Mobile Propulsion Systems, Simulation Sciences, Mechatronics, Thermodynamics, FEM

B.Tech Mechanical Engineering, JNTU Hyderabad

06/2019 – 04/2023

Hyderabad, India · CGPA 9.05 (GPA 1.3, German scale)

PUBLICATIONS

- "A Comprehensive Analysis of Two Innovative Aircraft Design Configurations," IJSED Research, 11/2022.
- Co-author, peer-reviewed paper on a rule-based FCEV operating-strategy control approach (Robert Bosch GmbH), in review at SAE.

LANGUAGES

English (Fluent, C1) · German (A2-B1) · Telugu / Hindi (Native)